

D-Vision-3D-Reconstructure

Memoir3D · Project Plan & Build Roadmap

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Project Overview

D-Vision-3D-Reconstructure (product name: Memoir3D) transforms crowd-sourced event photos — taken by multiple attendees with different phones, angles, lighting, and timestamps — into a single navigable 3D scene with a timeline scrubber. The user can fly around the venue in a browser and scrub through time to relive any moment of the party.

Market gap: No existing tool (Luma AI, Polycam, Scaniverse) handles crowd-sourced multi-attendee photos with a temporal layer. The product gap is real and the research foundation (WildGaussians NeurIPS 2024, CSS arXiv 2409.08562) is mature enough to build on.

Definitive Tech Stack

Backend (Python)

Package	Version	Role
FastAPI	0.115+	REST API + WebSocket server
ARQ	0.26+	Async Redis job queue (avoids MPS GPU context fork issues)
SQLModel	0.0.21+	ORM — single model for SQLite (local) + PostgreSQL (cloud)
Alembic	1.13+	Database migrations
redis[asyncio]	5.0+	Job queue + PubSub for live progress
Pillow + piexif	10+	Thumbnails, EXIF extraction

ML Pipeline (Apple Silicon Metal/MPS)

Tool	Role	Apple Silicon
COLMAP / pycolmap	Camera pose estimation (SfM)	✓ Metal-native (brew install)
hloc SuperPoint+SuperGlue	Feature matching for dark photos	✓ MPS supported
SAM2 (Meta)	Person masking per photo	✓ MPS with fallback env var
YOLOv8n (ultralytics)	Person bounding boxes for SAM2	✓ MPS supported

Depth Anything v2 Small	Depth hints (Apache 2.0)	✓ MPS + Core ML
OpenSplat	3D Gaussian Splatting training	✓ Metal-native build

Frontend (React / TypeScript)

Package	Role
React 18 + Vite 5	SPA — not Next.js (no SSR needed, simpler)
Three.js 0.169+	3D renderer
@mkkello/gaussian-splats-3d	Best maintained WebGL Gaussian Splatting renderer
Zustand 4.5+	Minimal state management
Tailwind CSS 4 + shadcn/ui	Styling + accessible components
axios + react-dropzone	File upload with progress events

ML Pipeline — Full Data Flow

All stages are chained sequentially. Each stage writes a checkpoint file. On failure or retry, completed stages are skipped automatically.

Step	Script	Time (300 photos)	Output
1 — Ingest & Validate	01_ingest.py	~5 sec	processed_images/, metadata.json
2 — Temporal Bucketing	02_bucket.py	<1 sec	buckets.json (15-min windows)
3 — COLMAP Full Event	03_colmap_full.py	20–90 min	sparse/0/ (camera poses)
4 — SAM2 Masking	04_mask.py	5–15 min	masks/*.png (255=exclude people)
5 — Window Sub-Recon	05_window_recon.py	<1 min	windows/{id}/sparse/0/
6 — OpenSplat Training	06_train_windows.py	~17 min/window	{bucket_id}.ply
7 — Post-Processing	07_postprocess.py	~2 min	scene.json + LOD preview.ply

Key COLMAP Settings for Party Photos

- **domain_size_pooling=True** — biggest win for dark/low-texture scenes
- **max_num_features=16384** — double default to survive noise
- **camera_mode=PER_IMAGE** — critical for mixed phones
- **hloc SuperPoint+SuperGlue** — replace SIFT for 2–3× more matched pairs

OpenSplat Key Parameters

- **max_gaussians=1,000,000** — hard cap to stay within 24 GB unified memory
- **sh_degree=3** — full spherical harmonics; reduce to 1 if OOM
- **iterations=30,000** — ~17 min on M-series Metal GPU
- **masks enabled** — pass SAM2 mask directory to exclude people from training

Phase-by-Phase Build Plan

Phase 0 — Proof of Concept	<i>Weeks</i> 1–3
• Prove the pipeline runs on your Mac mini BEFORE building anything.	
Phase 1 — Single-Event Pipeline	<i>Weeks</i> 4–8
• Python pipeline orchestrator: COLMAP→SAM2→OpenSplat as subprocess chain.	
Phase 2 — Browser 3D Viewer	<i>Weeks</i> 9–13
• React + Vite + Three.js scaffold. gaussian-splats-3d renderer.	
Phase 3 — Timeline Layer ■■	<i>Weeks</i> 14–20 (+4 buffer)
• RESEARCH WEEK 14: Read CSS paper + WildGaussians before coding.	
Phase 4 — Photo Collection UX	<i>Weeks</i> 21–26
• Event creation UI + shareable URL + QR code generation.	
Phase 5 — MVP Polish & Real Event	<i>Weeks</i> 27–32
• WebSocket live progress UI (6-stage pipeline). Email notifications.	
Phase 6 — Production Quality	<i>Weeks</i> 33–40
• SQLite → PostgreSQL migration (Alembic). Docker container.	

Milestones Summary

#	Milestone	Target Week	Key Success Criteria
M0	Environment working	3	All tools install; real benchmarks documented
M1	First reconstruction	2–3	Your room photos → viewable .ply
M2	Pipeline API	8	curl upload → .ply via API, handles errors

M3	Browser viewer	13	3DGS navigable in Chrome + Safari + Android
M4	Timeline scrubber	20 ±4	Scrubbing changes the 3D scene
M5	Multi-contributor UX	26	3 non-devs upload without help
M6	MVP real event	32	Real guests, no manual intervention
M7	Production quality	40	Cloud deployed, strangers use it

Critical Risks & Mitigations

Risk	P	I	Mitigation
COLMAP fails on dark/blurry photos	High	Critical	Blur rejection gate (Laplacian < 50), hloc SuperGlue matcher, domain_size_pooling
OpenSplat OOM (>24 GB)	High	Critical	max_gaussians=1M hard cap, memory monitor wrapper, auto-retry with sh_degree
OpenSplat Metal kernel missing	High	Critical	Validate on your Mac mini in Week 1 before Phase 1. Fallback: gsplat MPS fork
Phone clock drift ruins timeline	High	Critical	Visual anchor alignment, manual per-device offset slider in UI
No job queue → concurrent OOM	High	Critical	ARQ single-job queue in Phase 1. Never allow parallel GPU stages.
People ghost despite SAM2	High	Major	20px mask dilation, YOLO-prompted SAM2, float-Gaussian pruning post-train
WhatsApp strips EXIF	Med	Major	Warn user, manual time-range picker fallback
.ply too large for mobile	Med	Major	LOD preview.ply (<5 MB), chunked HTTP range streaming, mobile gets 100K Gaussian
Silent MPS CPU fallback	High	Major	Test with PYTORCH_ENABLE_MPS_FALLBACK=0 in CI to make fallbacks throw

Validation Gates

Before COLMAP:

- ≥ 30 images after blur filtering · ≤ 50% rejected for blur · ≥ 60% have DateTimeOriginal EXIF

Before OpenSplat:

- ≥ 70% images registered · ≥ 25 registered images · ≥ 5,000 sparse 3D points · reprojection error ≤ 2px

After OpenSplat:

- Validation PSNR ≥ 22 dB · SSIM ≥ 0.75 · Gaussian count ≤ 2,000,000 · .ply < 300 MB

Privacy & Legal Requirements

3D Gaussian splats rendering recognizable faces constitute biometric data under GDPR Article 9. These requirements must be implemented BEFORE any public launch.

Requirement	Implementation
Face blurring	InsightFace (MIT) detection + 25px Gaussian blur on texture layer. Auto-applied to all public scenes.
Data retention	Raw photos deleted 24h after reconstruction. COLMAP model: 30 days. Final .ply: 90 days default.
EXIF GPS stripping	Strip GPS from all uploads on ingest. Keep only DateTimeOriginal and FocalLength.
Right to erasure	'Report myself in this scene' button. Take-down within 72 hours.
Consent gate	Host confirms all attendees were informed before triggering reconstruction.
GDPR/CCPA	Privacy policy + data subject access request mechanism required before EU/CA users.

Library License Safety

Library	License	Commercial?
COLMAP / pycolmap	BSD-3-Clause	✓ Yes
SAM2 (Meta)	Apache 2.0	✓ Yes
Depth Anything v2 Small	Apache 2.0	✓ Yes
OpenSplat	AGPLv3	✓ Yes (AGPL-aware)
WildGaussians (reference)	MIT	✓ Yes
InstantSplat (NVLabs)	NVIDIA NC License	✗ NO — do not use
DUST3R / MAST3R	CC-BY-NC-SA 4.0	✗ NO — do not use
Depth Anything v2 Base/Large	CC-BY-NC-4.0	✗ NO — Small variant only

Key Papers to Read

#	Paper	Venue	Link	Why
1	3D Gaussian Splatting for Real-Time Radiance Field Rendering	Kerbl et al., SIGGRAPH	arxiv.org/abs/2308.04079	Foundation — everything builds on this
2	NeRF in the Wild: Neural Radiance Fields for Unconstrained Photo Collections	Martin-Brualla et al.,	arxiv.org/abs/2008.02268	Appearance embeddings for variable lighting — the core technique

3	CSS: Overcoming Pose and Scene Challenges in Crowd-Sourced 3D Gaussian Splatting	Chen et al., Sept 2022	arxiv.org/abs/2409.08562	YOUR EXACT PROBLEM — read before writing any code
4	WildGaussians: 3D Gaussian Splatting in the Wild	Kulhanek et al., NeurIPS 2024	arxiv.org/abs/2407.08447	MIT license — borrow appearance embedding approach
5	DUST3R: Geometric 3D Vision Made Easy	Wang et al., CVPR 2024	arxiv.org/abs/2412.14132	Pose-free pipeline (NC license but essential reading)
6	Gaussian in the Dark	2024	arxiv.org/abs/2408.09130	Dark venue / nightclub reconstruction